FLOODTASK

PROJECT REPORT

Forecast-Aware Reinforcement Learning for Multi-Pass Satellite Imaging of Evolving Floods

Project report - study region: Assam, India (Brahmaputra valley)

Semester 5 - Reinforcement Learning

Submitted by: _____

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Project details

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Study region	Assam, India (Brahmaputra valley); held-out test region
Course / semester	Semester 5 - Reinforcement Learning project
Submitted by	_____ Register no.: _____
Guide / faculty	_____
Department / institution	_____
Date	25 October 2026

Declaration

I declare that this report describes the project work carried out by me. The results reported here were produced by the code in the project repository and are read from the saved result files listed in each section. Simulated and synthetic components are identified as such wherever they are used.

Signature: _____ Date: _____

Abstract

Flood maps from synthetic aperture radar (SAR) satellites are essential during monsoon floods, but an agile SAR satellite can image only a small part of an affected region at each pass while the flood keeps moving. This project studies the resulting *tasking* problem as a partially observable Markov decision process: at each of 12 passes spaced 12 h apart, an agent selects 16 of the 256 tiles (720 m) of an 11.5 km scene, and a shared estimator fuses the delivered images into a flood map. The aim is a map that is both accurate and current. Training floods are simulated on real Copernicus DEM terrain with a HAND-based reach-storage model calibrated on training events, and image errors are emulated by a sensor model fitted to a U-Net trained on the Sen1Floods11 hand-labelled SAR data set.

A recurrent, forecast-aware actor-critic (CNN encoder, LSTM, FiLM, per-tile masked policy head; 0.77 M parameters) was trained with Proximal Policy Optimisation, together with four ablation variants, for 5 seeds each (25 runs). Evaluation followed a protocol frozen before testing, on 95 held-out scenes from three unseen flood events with Assam, India as the primary event. On these simulated test scenes the agent raised the mean per-pass IoU of the map to 0.268, against 0.225 for the best validation-tuned heuristic (+0.043, 95 % CI +0.035 to +0.051, Holm p = 2.1e-14), with a positive effect for all five seeds and all three events (India +0.040). However, neither memory nor forecast inputs added measurable value: a feed-forward agent was marginally better (-0.002) and removing forecasts changed nothing (-0.000). On real Sentinel-1 time series of the 2016 and 2022 Assam floods, the learned policy did not beat simple heuristics. The supported claim is therefore that learned tasking outperforms hand-built tasking in simulation; transfer to real events remains open. A 4-24 h ensemble flood outlook and a map viewer were also built. Decisions take 5.9 ms on one CPU thread, and every reported number regenerates from saved results with one command.

Keywords: reinforcement learning, PPO, satellite tasking, SAR flood mapping, POMDP, HAND, Assam, Brahmaputra

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Abbreviations

Abbreviation	Meaning	Abbreviation	Meaning
CI	Confidence interval	LSTM	Long short-term memory
CNN	Convolutional neural network	MDP	Markov decision process
D8	Eight-direction flow routing	MLP	Multi-layer perceptron
DEM	Digital elevation model	NWP	Numerical weather prediction
ERA5	ECMWF reanalysis, 5th generation	POMDP	Partially observable MDP
FiLM	Feature-wise linear modulation	PPO	Proximal policy optimisation
FN / FP	False negative / false positive	RL	Reinforcement learning
GAE	Generalised advantage estimation	RTC	Radiometric terrain correction
GRD	Ground range detected (SAR product)	SAR	Synthetic aperture radar
HAND	Height above nearest drainage	TD	Temporal difference
IoU	Intersection over union	VV / VH	SAR polarisations

1. Introduction

1.1 Background

Riverine floods in the Brahmaputra valley of Assam recur almost every monsoon season and affect large rural populations. Disaster-response agencies need flood-extent maps that are both correct and current: a map that was accurate three days ago can be misleading today, because water rises, spreads and recedes. During the monsoon, persistent cloud cover blocks optical satellites, which makes synthetic aperture radar the main source of flood observations. Calm open water reflects the radar signal away from the sensor and therefore appears dark, so it can be separated from land by thresholding or by learned segmentation models.

Commercial agile SAR satellites can be *tasked*: an operator requests particular target areas for each pass. Because each pass can only cover a limited area, the operator must decide where to look. The decision is sequential (what is observed now changes what is known later), it is made under uncertainty (the true flood is never observed directly), and it depends on many inputs: the age and confidence of existing observations, terrain, rainfall and rain forecasts. These properties make the problem a natural candidate for reinforcement learning.

1.2 Problem statement

Given a scene divided into 256 tiles, a fixed budget of 16 tile acquisitions at each of 12 satellite passes, random swath constraints and acquisition failures, select the tiles to image so that the flood map produced from the delivered images is as accurate as possible at every pass, using only information that is available at decision time.

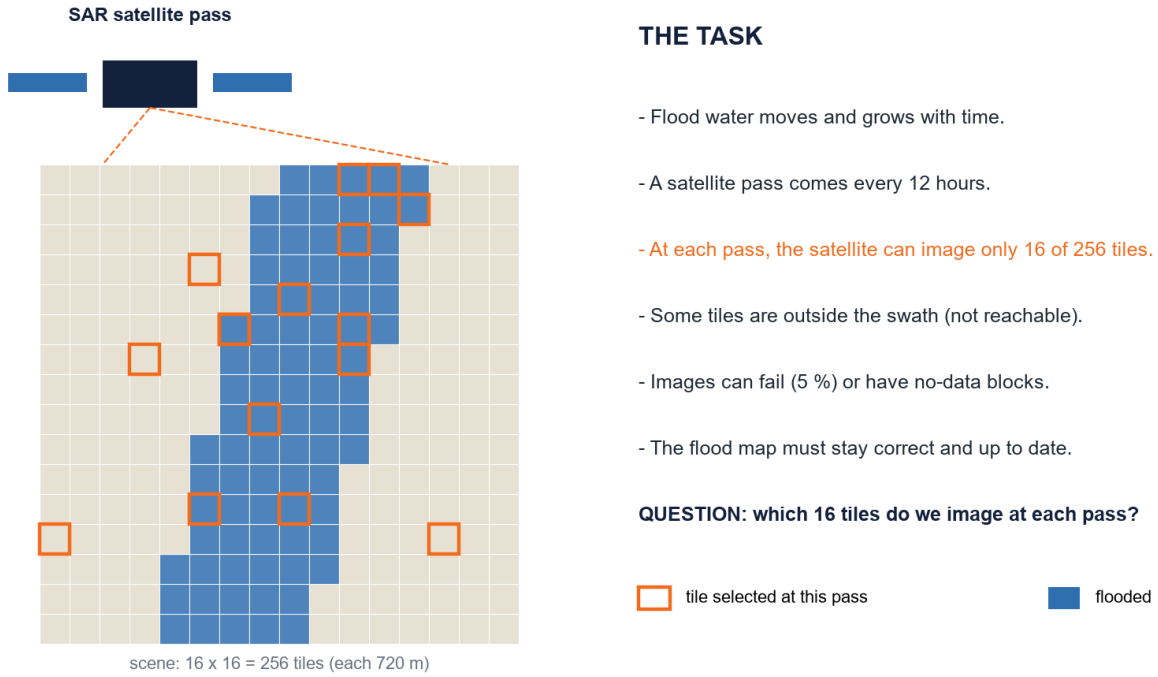


Figure 1.1 - The tasking problem: 16 of 256 tiles can be imaged at each pass.

1.3 Research question and hypothesis

RESEARCH QUESTION

Does a forecast-aware recurrent reinforcement learning agent produce more accurate flood maps than conventional tasking strategies under the same acquisition limits?

The confirmatory hypothesis H1, declared before testing, states that forecast-aware recurrent PPO achieves a higher mean per-pass IoU than the best-on-validation conventional baseline. It is tested against five conventional baselines and two PPO ablations (no memory, no forecasts) with Holm correction.

1.4 Objectives

1. Build a reproducible, leakage-free simulation environment for multi-pass flood-mapping satellite tasking on real terrain.
2. Train a SAR flood-segmentation model on real hand-labelled data and use its measured errors to model the sensor.
3. Train a forecast-aware recurrent PPO agent and four ablation variants with multiple seeds.
4. Compare the agents with validation-tuned heuristic baselines on held-out flood events, with India (Assam) as the primary test region.
5. Test robustness to forecast errors, missing data, budget, revisit interval, latency and sensor quality.
6. Replay real Sentinel-1 time series of the 2016 and 2022 Assam floods.
7. Report all results against requirements fixed in advance, including the ones that are not met.

1.5 Scope

The project is a research prototype. It does not connect to a real satellite operator, does not propagate real orbits and is not validated for operational or life-safety use. Flood dynamics for training are simulated; forecasts and gauges are synthetic in simulation (Section 3.5). These choices follow from the absence of open multi-date flood labels and archived forecasts for the study events.

1.6 Report organisation

Chapter 2 introduces the background concepts. Chapter 3 describes the data and study area. Chapter 4 presents the system design and the learning method. Chapter 5 covers the implementation. Chapter 6 gives the experimental protocol, Chapter 7 the results and Chapter 8 the discussion. Chapter 9 lists limitations and operational readiness, and Chapter 10 concludes.

2. Background

2.1 SAR flood mapping

Sentinel-1 provides C-band SAR images in VV and VH polarisation at about 10 m resolution. Open water produces low backscatter, and a global threshold such as Otsu's method on the VH band is a common baseline. Convolutional segmentation networks such as the U-Net [2] learn spatial context and are the standard learned approach. Sen1Floods11 [1] provides 446 hand-labelled 512 x 512 Sentinel-1 chips from 11 flood events and is the reference benchmark for this task.

2.2 Terrain and Height Above Nearest Drainage

HAND [7, 8] measures the height of each cell above the drainage cell into which it flows. It is computed from a DEM after depression filling [5] and D8 flow routing [6]. Low-HAND cells are hydrologically connected to the river and flood first, which makes HAND a strong, cheap predictor of riverine flood susceptibility and the basis of the simple flood model used here.

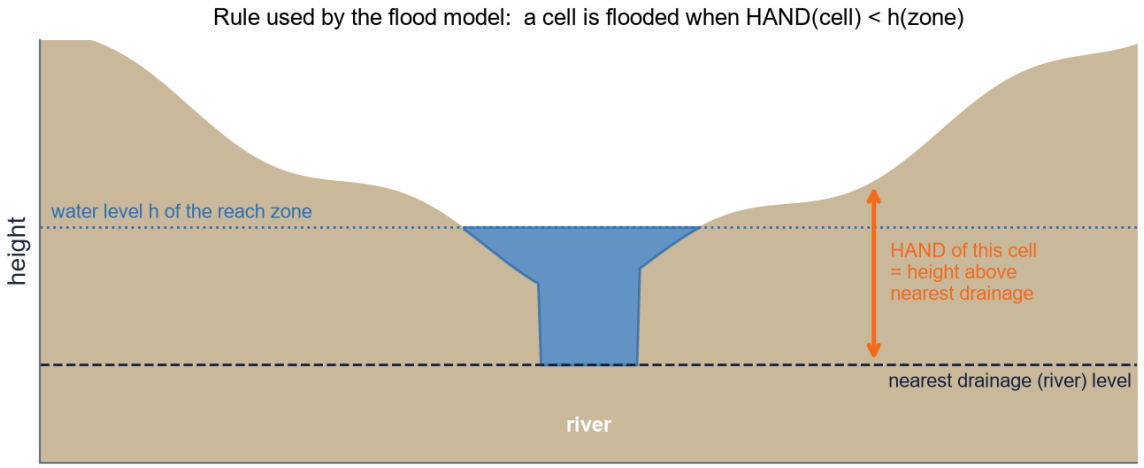


Figure 2.1 - HAND and the flooding rule used by the simulator.

2.3 Reinforcement learning

In reinforcement learning an agent interacts with an environment: at step t it observes the state s_t, takes an action a_t according to a policy π, and receives a reward r_{t+1}. The goal is to maximise the expected discounted return. When the agent receives only an observation o_t that does not reveal the full state, the problem is a POMDP, and agents typically need memory such as an LSTM [4].

$$G_t = r_{t+1} + \gamma r_{t+2} + \gamma^2 r_{t+3} + \ldots = \sum_{k=0}^{\infty} \gamma^k r_{t+k+1}$$

$$V^{\pi}(s) = \mathbb{E}_{\pi}\left[r_{t+1} + \gamma V^{\pi}(s_{t+1}) \mid s_t = s \right]$$

Actor-critic methods learn a policy (actor) and a value function (critic) together; the critic provides the advantage estimate that weights the policy gradient. PPO [3] stabilises policy-gradient training by clipping the probability ratio between the new and old policy, and GAE [9] provides a low-variance advantage estimate.

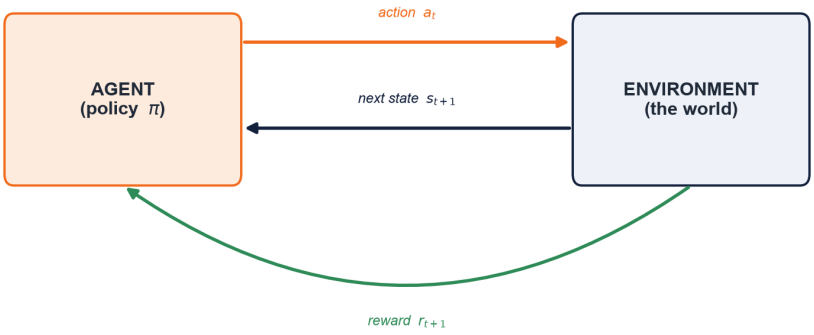


Figure 2.2 - The agent-environment interaction loop.

2.4 Satellite tasking

Satellite tasking is classically solved with heuristics or combinatorial optimisation that maximise the value of requested targets subject to agility, swath and downlink constraints. For an evolving phenomenon such as a flood, the value of an image depends on what is already known and how old it is, which motivates sequential, information-driven policies. This project compares learned policies against five such heuristics, including a greedy expected-error rule that combines terrain susceptibility with observation uncertainty and staleness.

3. Data and study area

3.1 Study area

The primary test region is the Brahmaputra valley of Assam, India, covered by the 68 Sen1Floods11 'India' chips at 92.4-94.0 E, 25.7-27.3 N (Sentinel-1 date 2016-08-12). The region was chosen at the project owner's request before any model was trained, and India was moved from the training set to the primary test set. It is never used for training, tuning or model selection.

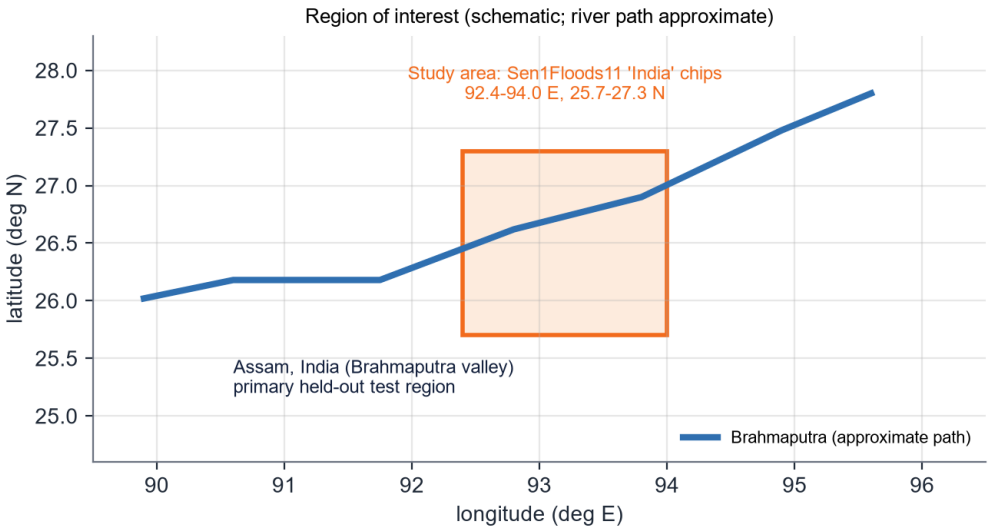


Figure 3.1 - Study area (schematic).

3.2 Data sources

#	Source	Status	Resolution / period	Use
D1	Sen1Floods11 v1.1 hand-labelled (S1, labels, Otsu labels)	Observation + manual labels	10 m chips; one date per event	U-Net training and test; calibration
D2	Copernicus DEM GLO-30	Observation-derived	30 m, resampled (mean) to 60 m	HAND, drainage, reach zones
D3	ERA5 daily precipitation (Open-Meteo archive)	Reanalysis	0.25 deg, daily	Storm-size range; replay rain
D4	Sentinel-1 RTC (Microsoft Planetary Computer)	Observation	10 m; Assam 2016 (11 dates), 2022 (25 dates)	Real-event replay
D5	Simulated flood trajectories	Synthetic	60 m, hourly	Training and controlled tests
D6	Synthetic forecasts, gauges, rain observations	Synthetic	Per tile, per pass	Agent inputs; robustness tests

All raw downloads are recorded with SHA-256 checksums (2,230 Sen1Floods11 files) and derived files in results/data_manifest.json. No archived numerical weather forecast is used anywhere: TIGGE requires registration, and the Open-Meteo previous-runs archive returned no usable precipitation for the 2022 Assam period.

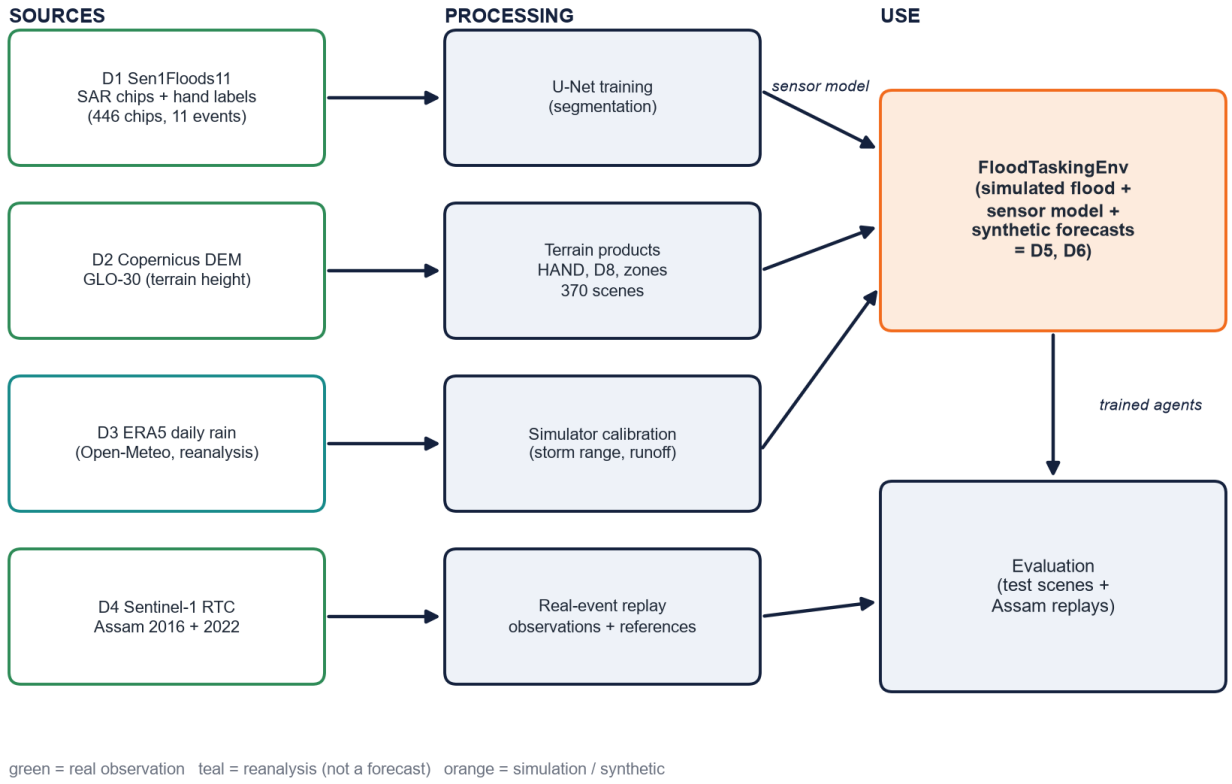


Figure 3.2 - Data flow from sources to the environment and evaluation.

3.3 Event-level split

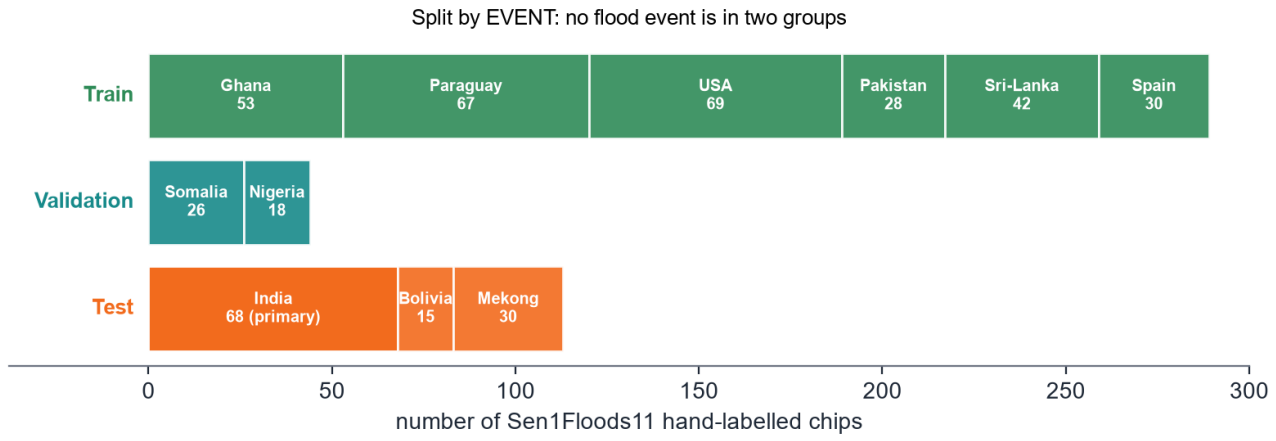


Figure 3.3 - Event-level split of the Sen1Floods11 chips.

Splitting by event prevents near-duplicate imagery of one flood from appearing in both training and test data. Validation events are used for checkpoint selection, calibration and baseline tuning only.

3.4 Scenes

For each chip location, a DEM window is processed (depression filling, D8 directions, flow accumulation, drainage threshold of 1,000 cells chosen by training-event AUC, HAND, reach zones of 48-cell blocks with acyclic outlet links). This yields 370 scenes of 192 x 192 cells at 60 m, of which 59 are India test scenes, 13 Bolivia and 23 Mekong.

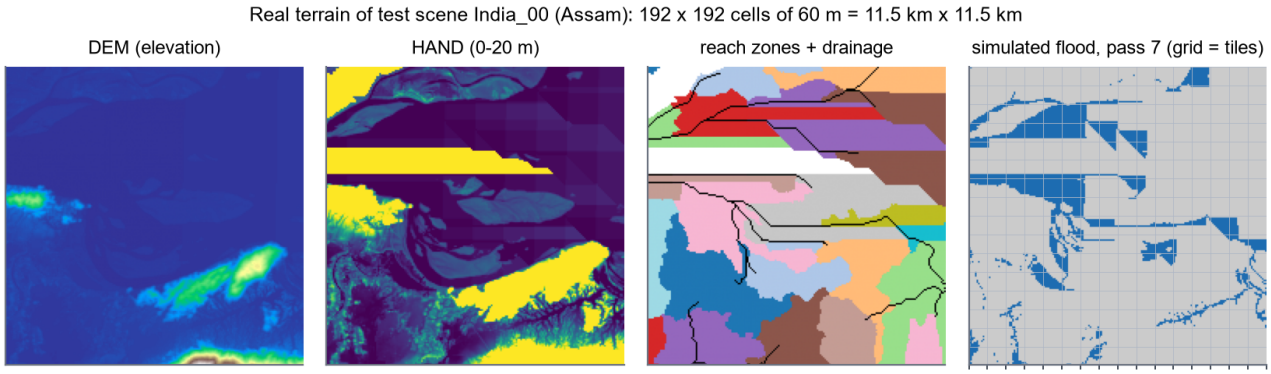


Figure 3.4 - Terrain products of test scene India_00 and one simulated flood state.

3.5 Real versus simulated components

Component	Status
SAR segmentation (U-Net)	Trained and tested on real Sen1Floods11 chips with hand labels, split by event
Terrain (HAND, drainage, reach zones)	Real Copernicus DEM
Flood evolution for training and controlled tests	Simulated: HAND reach-storage model with synthetic storms scaled to ERA5 totals
Forecasts and gauges	Synthetic with documented error models (simulation); ERA5 pseudo-forecast and missing gauges (replay)
Real-event test	Real Sentinel-1 RTC series over Assam; reference masks are model-derived

4. System design and methodology

4.1 Architecture

The system consists of a terrain and flood simulator, a sensor model, the tasking environment with a shared map estimator, the policies (learned and heuristic), and an evaluation pipeline. Policies receive only the observation; ground truth enters only the training reward, the evaluator metrics and the non-operational oracle.

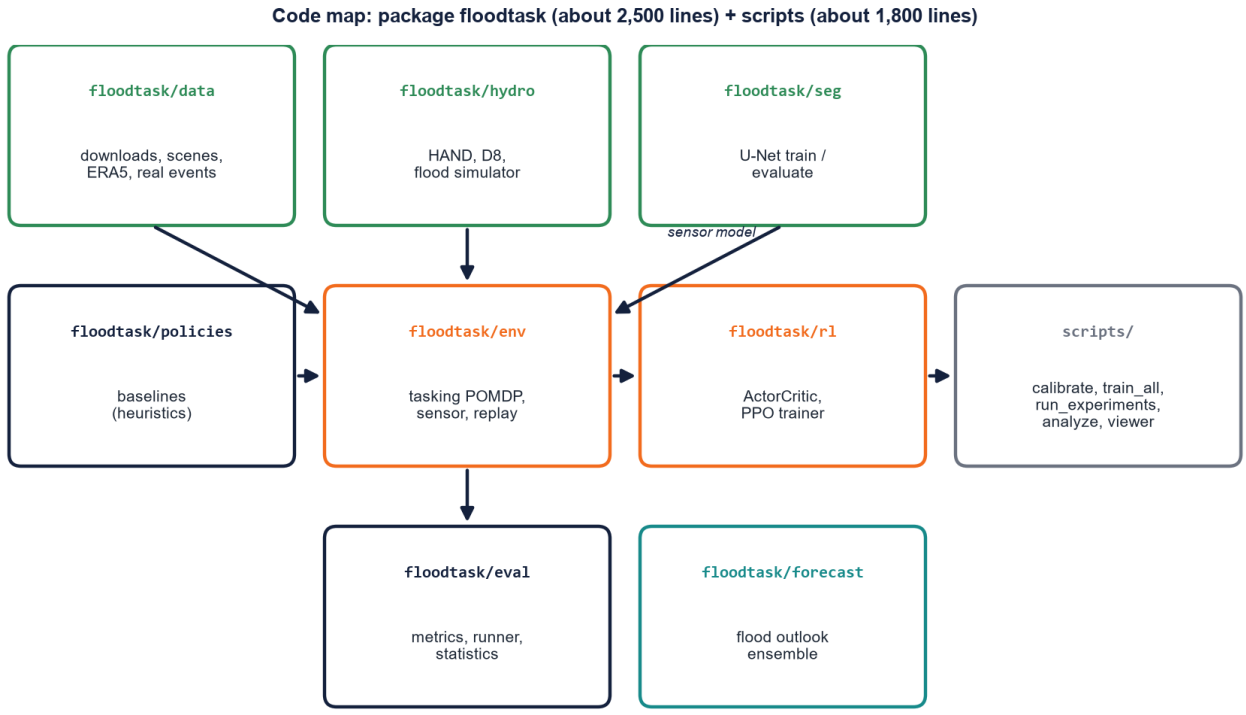


Figure 4.1 - Software modules and their dependencies.

4.2 Flood simulator

The simulator is a conceptual HAND reach-storage model, not a hydrodynamic model. One to four Gaussian rain cells move over an extended grid; totals are rescaled to a scene-mean storm total drawn from an ERA5-informed range (27-216 mm). Each reach zone is a linear reservoir; outflow is routed to the downstream zone within the same hour, and an external upstream inflow follows a gamma unit hydrograph with a 12-60 h lag. Storage above an overbank threshold is converted into a water level by the zone's storage-volume curve, and a cell floods when its HAND is below the zone level.

$$S_z(t+1) = \left[(I-U)^{-1}(S(t)+q(t))\right]_z\left(1-\tfrac{1}{r_z}\right)$$

$$F_t(x) = \mathbb{1}\left[\mathrm{HAND}(x) < h_{z(x)}(t)\right]$$

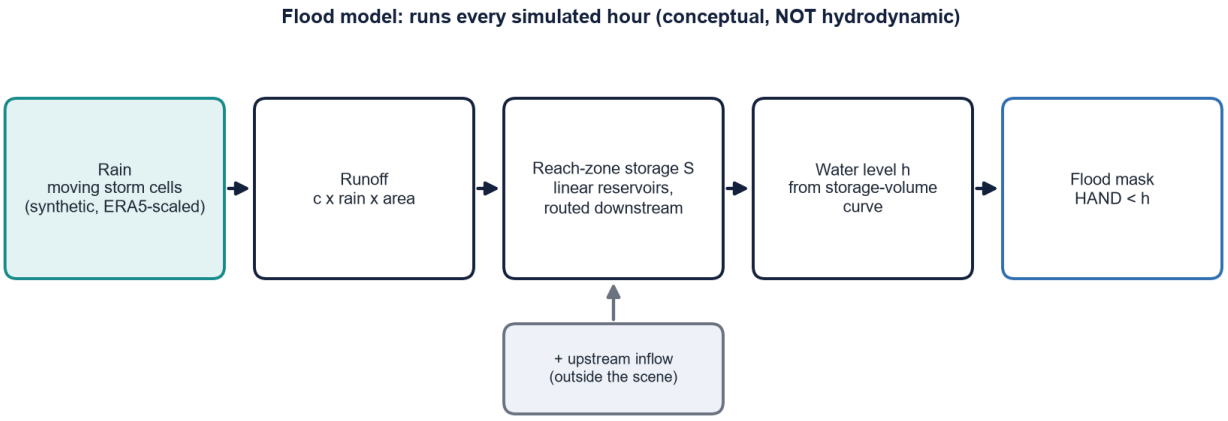


Figure 4.2 - Hourly simulation chain.

Runoff coefficient, external basin scale and the overbank storage range were calibrated on training events only so that the distribution of simulated flooded fractions matches the observed fractions of training chips (Kolmogorov-Smirnov statistic 0.13). Residence times (12-72 h) and lags remain assumptions because no multi-date labels exist.

4.3 Segmentation model and sensor model

A U-Net (4 levels, 32-512 channels) takes the VV and VH bands and outputs a calibrated water probability. It is trained with weighted cross-entropy plus Dice loss, AdamW (lr 1e-3, cosine schedule), 80 epochs, mixed precision and three seeds; the best validation epoch is kept and a temperature is fitted on validation pixels.

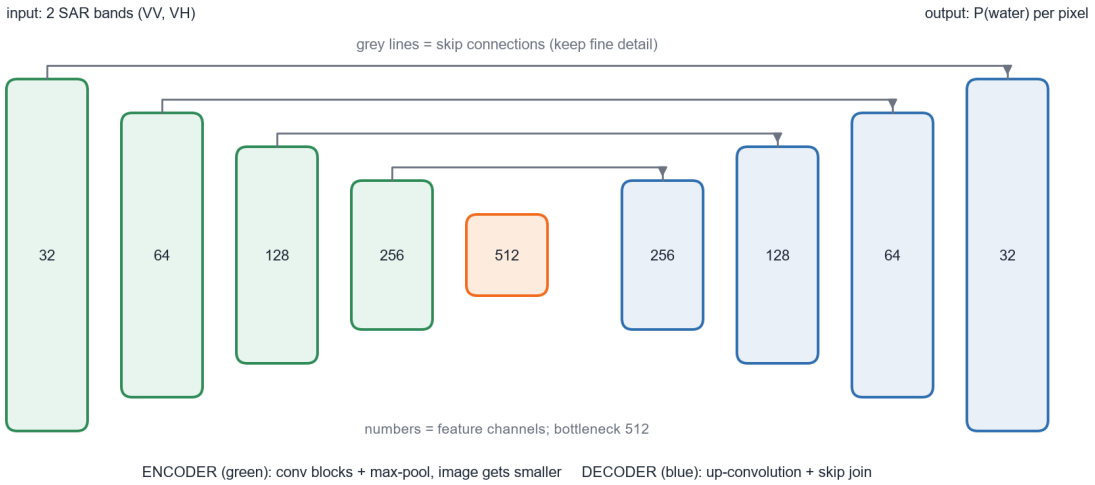


Figure 4.3 - U-Net used for SAR flood segmentation.

The environment does not run the U-Net on synthetic images. Instead, an acquired tile returns a probability map from a parametric sensor model whose parameters were fitted to the U-Net's errors on validation events at 60 m (a = 3, β = -1.0, σ_n = 2.0, blur 1.0, partial no-data probability 0.024):

$$\hat{\pi} = \sigma(a\,(2\,G_b * F - 1) + \beta + \sigma_n\,\varepsilon_{corr})$$

4.4 POMDP formulation

Element	Definition
Hidden state	Reach storages and levels, flood masks, full rain field (past and future), episode physics, pre-sampled swaths, failures, noise seeds and forecast errors
Action	One tile index a in {0, ..., 255}; must lie in the pass swath and not be selected earlier in the pass
Step / pass / episode	1 step = 1 tile; 16 steps = 1 pass; 12 passes (12 h apart, after 48 h spin-up) = 192 steps
Observation	18 x 16 x 16 grid (map estimate, uncertainty, age, data quality, forecasts, rain, terrain, constraints) + 10-element vector + action mask
Transition	Actions never change the flood, only the knowledge of it. Images are integrated after the 16th step; the pass then advances
Reward	Cost-weighted map-error reduction in the selected tile at delivery time (missed flood weight 3)
Termination	Fixed horizon of 12 passes for every policy

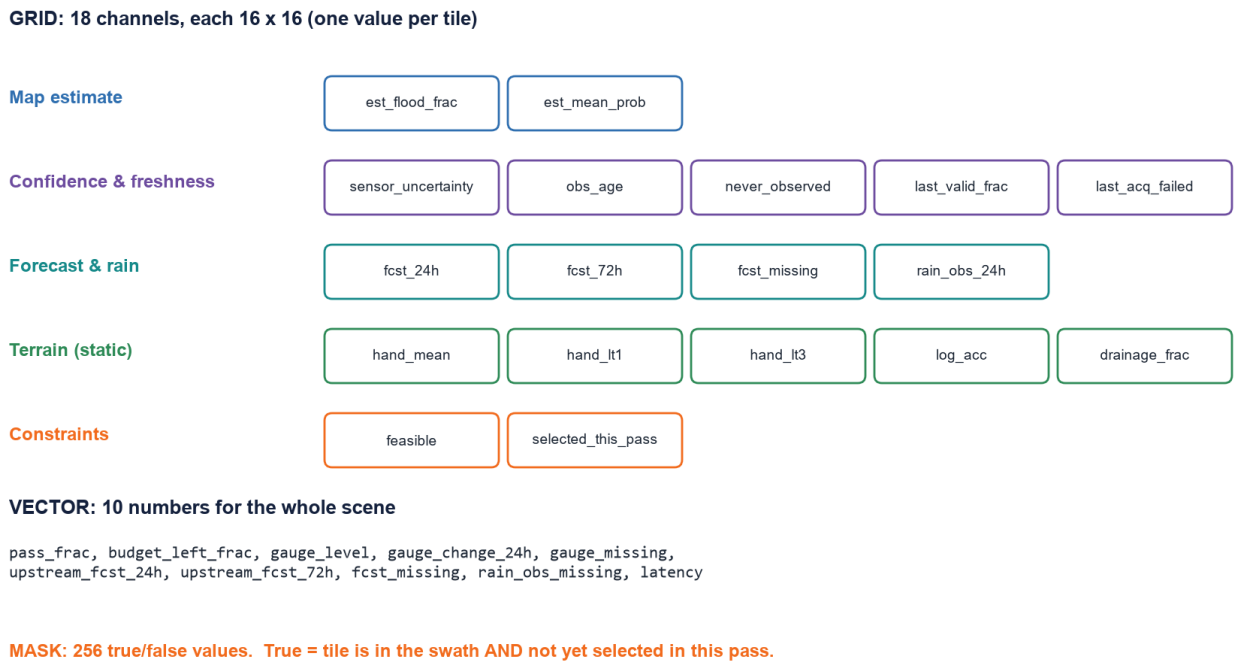


Figure 4.4 - Observation channels.

The map uncertainty that drives several baselines separates confidence from staleness: H_i is the mean binary entropy of the last observation of tile i and age_i the time since it was delivered.

$$u_i = 1 - (1 - H_i)\, e^{-\mathrm{age}_i/\tau}$$

4.5 Shared map estimator

All policies feed the same estimator, so they differ only in which tiles they select. The estimator keeps the latest valid probability per cell, fused in logit space with a terrain prior P(water | HAND) fitted on training events; no-data cells keep their previous value and cells with probability ≥ 0.5 are mapped as flooded. The prior was adopted because it improved validation IoU for every policy.

$$\text{logit}\,P(\text{water} \mid \text{HAND}) = -1.338 - 0.731\ln(1+\text{HAND})$$

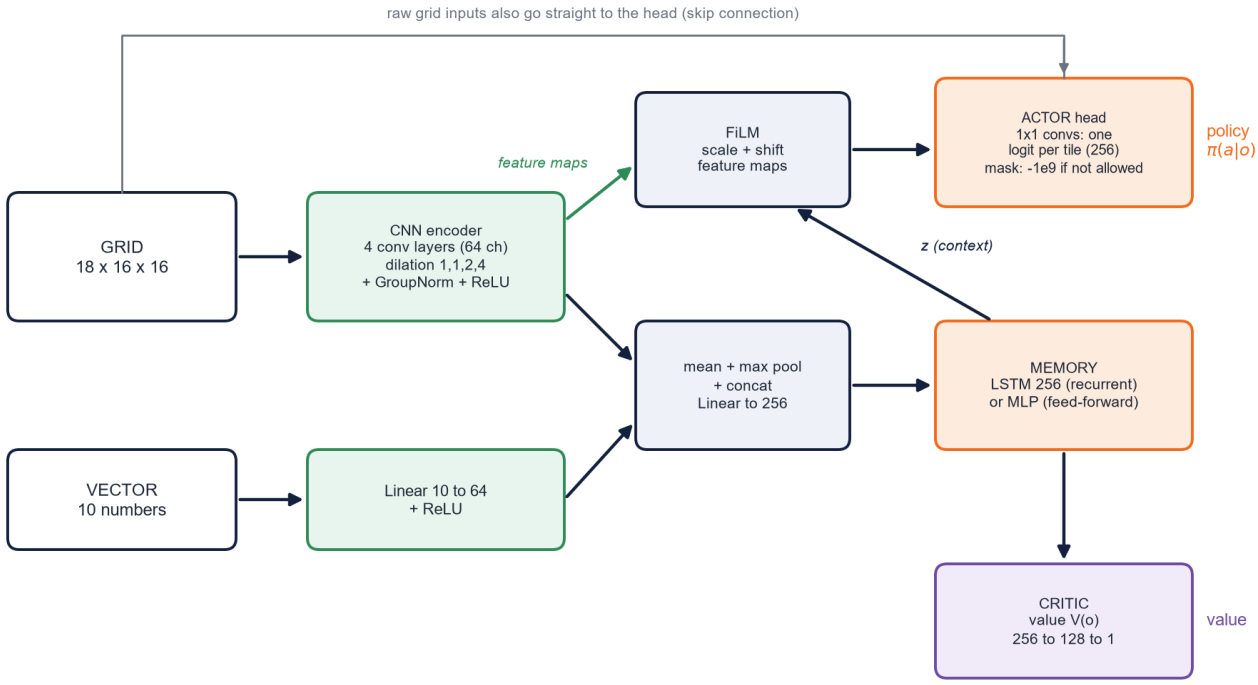
4.6 Reward design

$$r = \frac{1}{144}\sum_{x \in \text{tile}} ([3\,FN^{-}(x) + FP^{-}(x)] - [3\,FN^{+}(x) + FP^{+}(x)])$$

The difference reward credits each selection with the change in cost-weighted error of its own tile; with zero latency the per-tile rewards of a pass sum to the global error reduction (unit-tested). The missed-flood weight was chosen on validation by requirement R12, the Spearman correlation between episode return and the evaluation endpoint:

Reward candidate	Spearman (episode level)	Spearman (policy level)	Decision
Error reduction, FN weight 1	0.77	0.66	Rejected
Error reduction, FN weight 3	0.83	1.00	Selected
IoU difference reward	0.49	-	Rejected

4.7 Actor-critic network



Size: 769,474 parameters (recurrent), 308,930 (feed-forward). One decision on 1 CPU thread: 5.9 ms mean.

Same network for every PPO variant; only the memory block, the inputs and the reward change.

Figure 4.5 - Actor-critic network for per-tile tasking.

A four-layer dilated CNN with GroupNorm encodes the grid; mean and max pooling plus a vector encoder feed a 256-unit LSTM (or an MLP in the feed-forward variant). Its output z modulates the spatial features through FiLM and feeds the critic. The actor head combines the modulated features with the raw grid through

1 x 1 convolutions to produce one logit per tile; infeasible tiles receive a logit of -10^9. GroupNorm proved necessary: without it the activations shrank about 2.5 times per layer and all tile logits were nearly equal.

$$f' = f \odot (1 + \gamma(z)) + \beta(z)$$

$$\pi(a \mid o) = \frac{e^{\ell_a}}{\sum_{b \text{ allowed}} e^{\ell_b}}, \qquad \ell_b = -10^9 \text{ if not allowed}$$

4.8 Proximal policy optimisation

$$\hat{A}_t = \delta_t + (\gamma\lambda)\,\delta_{t+1} + (\gamma\lambda)^2\,\delta_{t+2} + \ldots \qquad (\gamma = 0.9,\ \lambda = 0.95)$$

$$L^{CLIP}(\theta) = \mathbb{E}_t[\min(r_t(\theta)A_t,\ \text{clip}(r_t(\theta), 1-\varepsilon, 1+\varepsilon)A_t)]$$

$$L = -L^{CLIP} + 0.5\,(V_\theta(o_t) - \hat{G}_t)^2 - 0.003\,H[\pi_\theta(\cdot \mid o_t)]$$

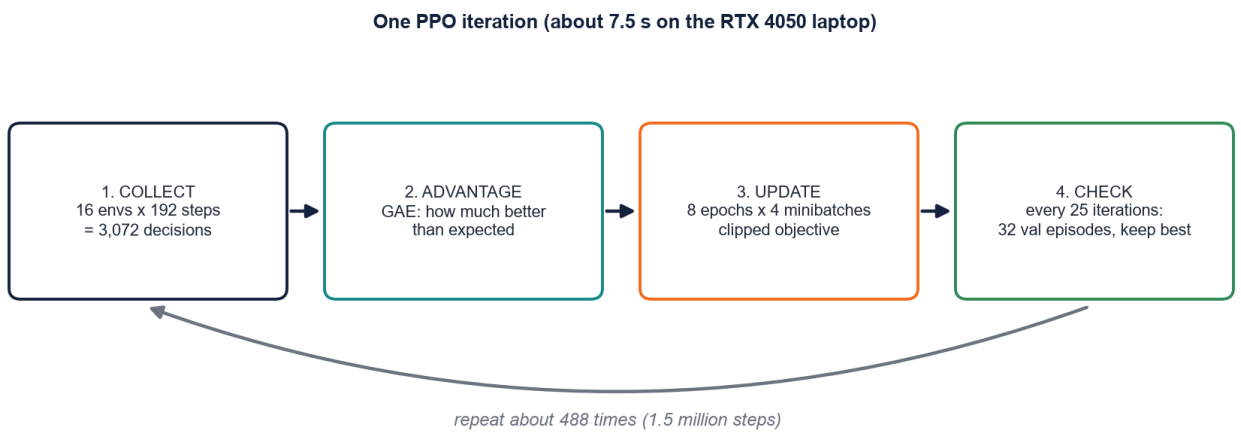


Figure 4.6 - One PPO iteration.

Hyperparameter	Value	Hyperparameter	Value
Total steps	1.5 M	γ (discount)	0.9
Parallel environments	16	GAE λ	0.95
Learning rate	1e-3, linear decay	Clip ε	0.2
Epochs per iteration	8	Entropy coefficient	0.003
Minibatches	4 (whole sequences)	Value coefficient	0.5
Gradient clip	0.5	Checkpoint	Best validation IoU, 32 episodes / 25 iterations

γ, learning rate, entropy coefficient and epochs were selected by a five-configuration validation sweep of feed-forward PPO (400 k steps each); the chosen setting reached a validation mean IoU of 0.226. Recurrent PPO was not tuned separately, which is conservative for H1.

4.9 Baseline policies

Baseline	Tile score	Tuning
random	Uniform random	-
stalest	Observation age (uniform coverage)	-
flooded_first	Mapped flood fraction + w_t · low-terrain fraction (never observed) + w_a · age	Validation grid search
forecast	(Forecast 24 h + 0.5 · forecast 72 h) · susceptibility + w_a · age	Validation grid search
greedy_unc	Susceptibility (terrain, flood estimate, forecast rain) × uncertainty u	Validation grid search (48 episodes)
oracle	True current map error of the tile (non-operational)	-

4.10 Variants

Variant	Memory	Forecasts	Reward	Role
recurrent	LSTM	Yes	Error reduction (FN weight 3)	Main method
feedforward	MLP	Yes	Same	Value of memory
recurrent_noforecast	LSTM	No	Same	Value of forecasts
recurrent_unc	LSTM	Yes	+ 0.1 uncertainty reduction	Ablation
recurrent_iou	LSTM	Yes	Pass-level IoU change	Ablation

5. Implementation

5.1 Software and hardware

Item	Value
Language and libraries	Python 3.12.6, PyTorch 2.5.1 + CUDA 12.4, NumPy, SciPy, rasterio, pystac-client, statsmodels, matplotlib
Hardware	Windows 11 laptop, NVIDIA RTX 4050 Laptop GPU (6 GB), 16 logical CPUs, 24 GB RAM
Code size	Package floodtask about 2,500 lines; scripts about 1,800 lines of Python
Configuration	configs/data.yaml, env.yaml, ppo.yaml, split.yaml
Tests	pytest: HAND, metrics, environment leakage, feasibility, reward decomposition, statistics

5.2 Pipeline

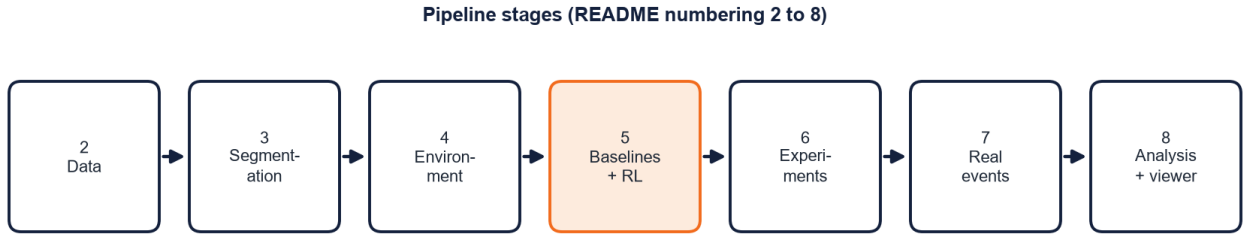


Figure 5.1 - Pipeline stages.

```
python -m floodtask.data.sen1floods11 --verify   # data + checksums
python -m floodtask.data.scenes                   # DEM, HAND, 370 scenes
python -m floodtask.seg.train --seed {0,1,2}      # U-Net
python scripts/fit_sensor_model.py                # sensor model (validation events)
python scripts/calibrate_sim.py                   # simulator calibration (training events)
python scripts/tune_baselines.py                  # baselines (validation)
python scripts/train_all.py                       # 5 variants x 5 seeds
python scripts/run_experiments.py --suite all     # test suites
python scripts/run_real_event.py --event assam2016   # and assam2022
python scripts/analyze.py && python scripts/make_manifest.py --verify
```

5.3 Information boundaries and testing

- test_no_future_truth_leakage corrupts all hidden future state and asserts that the observation is unchanged.
- Infeasible actions raise an exception in the environment; 6,080 core-suite episodes raised none.
- With zero latency, per-tile rewards of a pass sum exactly to the global error reduction (unit test).
- Every EpisodeSpec fixes scene, seed, physics, storms, swaths, failures, sensor noise and forecast errors, so all policies are compared on identical episodes.

5.4 Training execution



Figure 5.2 - Training matrix: 25 independent runs of 1.5 M steps.

The 25 runs were planned across two machines (configs/split.yaml). The second machine failed (no GPU, and the Colab attempt could not start the training script), so its 13 runs were executed on the project laptop with unchanged settings and seeds. Run times were 46-103 minutes; one run (recurrent_iou seed 0) needed one re-run.

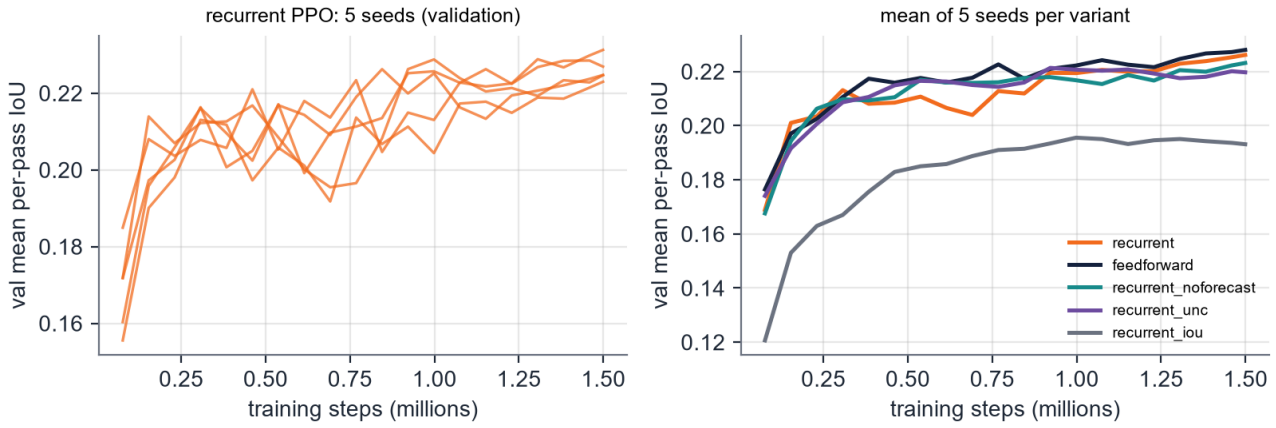


Figure 5.3 - Validation mean per-pass IoU during training (from results/rl//seed*/train_log.json).*

5.5 Reproducibility

Seeds are fixed and recorded, and every result file carries software versions, GPU and timestamp. GPU training is not bit-deterministic (cuDNN, fp16), but evaluation of saved checkpoints is deterministic. python scripts/analyze.py regenerates every table and figure; python scripts/make_manifest.py --verify checks 678 files with 0 mismatches.

5.6 Map viewer

A single-page web application (results/viewer) with five tabs - Flood maps, Results, Robustness, Real events and Training - shows the PPO and greedy-uncertainty maps on the Assam replays and on simulated Bolivia, India and Mekong test scenes, with the 4/6/12/24 h outlook panel. It is built by scripts/build_viewer_data.py and scripts/export_results_data.py.

6. Experimental protocol

The protocol (docs/experimental_protocol.md) was written before any test-event evaluation of tasking policies and frozen with a timestamp and SHA-256 hash (results/protocol_freeze.json). Deviations made before testing on validation evidence only (endpoint restriction, reward weight, estimator prior, simulator thresholds) are listed in the protocol.

6.1 Primary endpoint

$$\mathrm{IoU} = \frac{TP}{TP + FP + FN}$$

$$\overline{\mathrm{IoU}} = \frac{1}{|\mathcal{P}_{flood}|} \sum_{p \in \mathcal{P}_{flood}} \mathrm{IoU}_p$$

The primary endpoint is the mean IoU over flood-present passes (true flooded fraction ≥ 0.1 %), evaluated 3 h after each pass. It rewards both accuracy and timeliness. Secondary endpoints are final-pass IoU, F1, precision, recall, area bias, boundary F1, time to IoU 0.5 and coverage.

6.2 Statistics

- Test set: every test scene x 4 episode seeds (95 scenes, 378 scored episodes per policy); RL metrics averaged over 5 training seeds.
- Analysis unit: scene (paired differences averaged within scene). 95 % CI by bootstrap over scenes, stratified by event, 10,000 resamples.
- Test: Wilcoxon signed-rank on scene-level differences; Holm correction over the 7 confirmatory comparisons; effect size Cohen's d_z.
- Cross-check: linear mixed-effects model on episodes with a random intercept per scene.
- India is reported separately as the primary regional result.

6.3 Experiment suites

Suite	Conditions
Core	Operational setting: K = 16, 12 h revisit, latency 0, operational forecast mix
Ablation	No forecast, feed-forward, uncertainty bonus, IoU reward
Forecast robustness	Accurate, noisy, biased low (x0.4), biased high (x2.5), displaced (4-6 tiles), uninformative, missing; missing rain and gauge
Budget / revisit / latency	8, 16, 32 tiles; 6, 12, 24 h; latency 0 or 1 pass
Sensor quality	Perfect, calibrated, degraded
Real events	Assam 2016 (11 dates) and 2022 (25 dates) replays, 33 scenes each, descriptive only

7. Results

7.1 Segmentation

Model (test events)	Pooled IoU	Bolivia	India	Mekong
U-Net seed 0	0.667	0.687	0.547	0.812
U-Net seed 1	0.610	0.588	0.502	0.756
U-Net seed 2	0.678	0.611	0.540	0.863
U-Net mean (SD)	0.652 (0.036)	-	0.530	-
Otsu reference	0.650	0.697	0.529	-

The U-Net meets requirement R11 (≥ Otsu on pooled test events) by a small margin; seed 1 alone is below Otsu. India is the hardest event. Seed 0 is the deployed model for real-event observations.

7.2 Core comparison

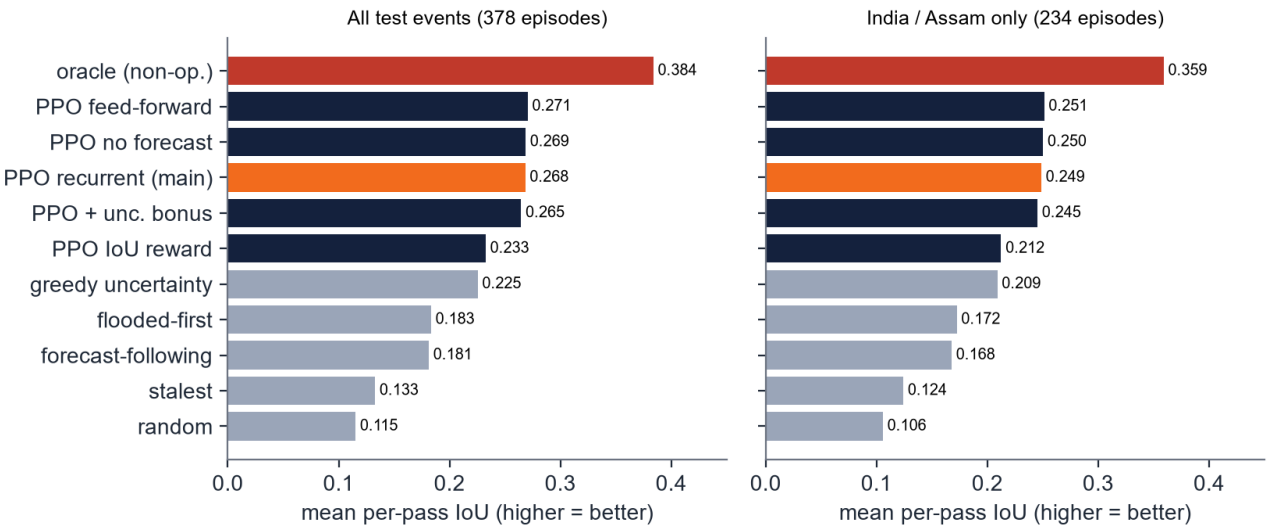


Figure 7.1 - Mean per-pass IoU of all policies on the held-out test scenes.

Policy	Mean per-pass IoU	Final IoU	Final F1	Reached IoU 0.5
oracle (non-operational)	0.384	0.466	0.624	47.4 %
PPO feed-forward	0.271	0.355	0.510	18.3 %
PPO recurrent, no forecast	0.269	0.353	0.508	17.3 %
PPO recurrent (main)	**0.268**	**0.354**	**0.509**	**17.7 %**
PPO recurrent + uncertainty bonus	0.265	0.353	0.507	18.6 %
PPO recurrent, IoU reward	0.233	0.313	0.462	9.8 %
greedy_unc	0.225	0.303	0.452	4.2 %
flooded_first	0.183	0.235	0.365	1.9 %
forecast	0.181	0.237	0.368	2.4 %
stalest	0.133	0.222	0.351	0.8 %
random	0.115	0.183	0.300	0.0 %

7.3 Confirmatory comparisons (H1)

PPO recurrent minus	Diff.	95 % CI	Wilcoxon p	Holm p	d_z	Seeds > 0	Bolivia / India / Mekong
random	+0.153	[+0.144, +0.162]	2.6e-17	1.8e-16	3.13	5/5	+0.130 / +0.143 / +0.192
stalest	+0.135	[+0.127, +0.143]	2.6e-17	1.8e-16	2.87	5/5	+0.111 / +0.124 / +0.177
flooded_first	+0.085	[+0.076, +0.094]	2.6e-17	1.8e-16	1.75	5/5	+0.065 / +0.076 / +0.118
forecast	+0.087	[+0.079, +0.095]	2.6e-17	1.8e-16	2.22	5/5	+0.086 / +0.081 / +0.103
greedy_unc	+0.043	[+0.035, +0.051]	7.0e-15	2.1e-14	1.11	5/5	+0.058 / +0.040 / +0.042
PPO feed-forward	-0.002	[-0.004, -0.000]	0.0057	0.011	-0.22	1/5	-0.006 / -0.003 / +0.001
PPO no forecast	-0.000	[-0.003, +0.002]	0.40	0.40	-0.03	3/5	-0.004 / -0.002 / +0.005

H1 is supported against all five conventional baselines and not supported against the two PPO ablations. The mixed-effects cross-check agrees in sign and significance for all seven comparisons (e.g. vs greedy_unc +0.047, p = 9.1e-23; vs feed-forward -0.0025, p = 0.048; vs no forecast -0.0001, p = 0.96).

7.4 India (Assam) test scenes

PPO recurrent minus	Diff.	95 % CI	Holm p	Seeds > 0
random	+0.143	[+0.133, +0.153]	1.7e-10	5/5
stalest	+0.124	[+0.115, +0.134]	1.7e-10	5/5
flooded_first	+0.076	[+0.067, +0.085]	1.7e-10	5/5
forecast	+0.081	[+0.072, +0.090]	1.7e-10	5/5
greedy_unc	+0.040	[+0.031, +0.049]	1.1e-9	5/5
PPO feed-forward	-0.003	[-0.005, -0.001]	0.03	1/5
PPO no forecast	-0.002	[-0.004, +0.001]	0.16	1/5

On the 59 India scenes (234 episodes), PPO reaches 0.249 mean per-pass IoU against 0.209 for greedy_unc; the final-pass area bias is +0.12, within the 0.25 limit.

7.5 Map quality over time

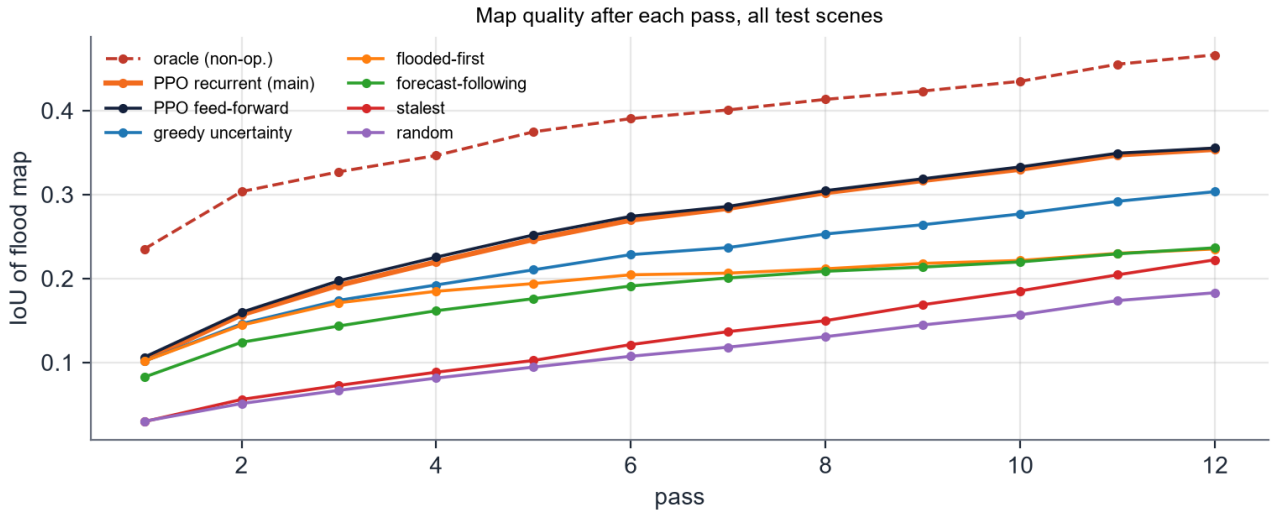


Figure 7.2 - IoU after each pass (flood-present passes), all test scenes.

The PPO advantage appears from the first passes and widens over the episode. The oracle bound (0.384) shows that substantial headroom remains even with the same estimator and budget.

7.6 Flood-size strata

Policy	Small floods	Medium floods	Large floods
PPO recurrent	0.200	0.273	0.333
PPO feed-forward	0.198	0.275	0.338
greedy_unc	0.165	0.229	0.283
flooded_first	0.114	0.181	0.255
oracle	0.313	0.376	0.462

7.7 Robustness and operating conditions

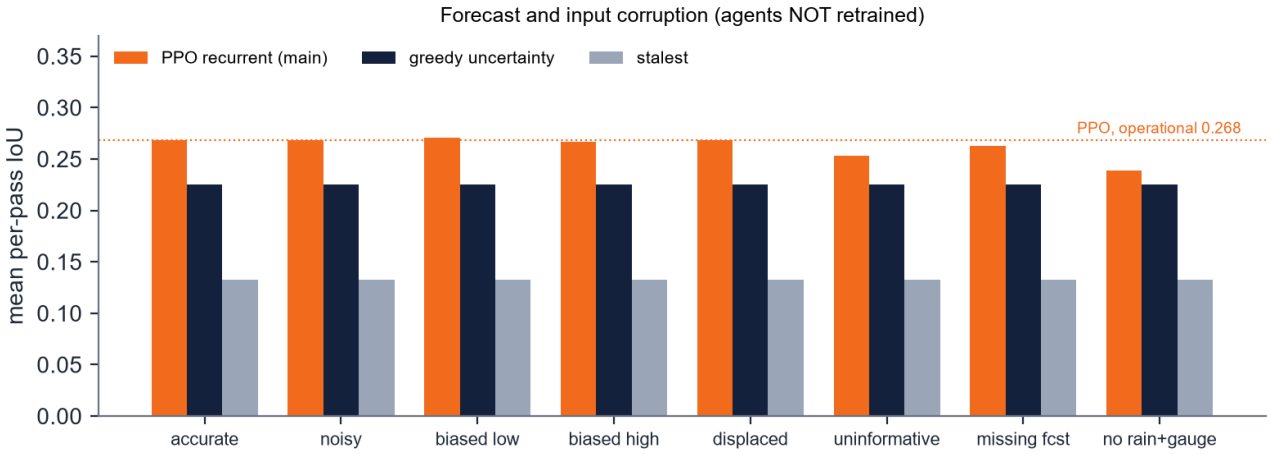


Figure 7.3 - Forecast and input corruption; agents are not retrained.

Relative to the operational forecast setting (0.268), PPO changes by -0.1 % (noisy), -0.7 % (biased high), +0.8 % (biased low), +0.1 % (displaced), -5.7 % (uninformative) and -2.2 % (missing forecasts); with missing forecasts PPO (0.263) remains far above stalest (0.133). Removing rain and gauge observations costs 11.0 %, above the 10 % limit of requirement R6.

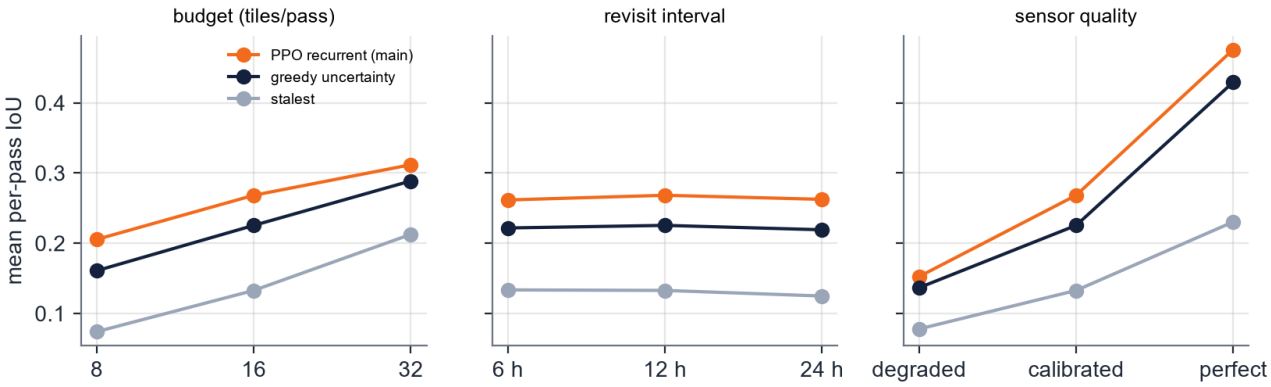


Figure 7.4 - Budget, revisit interval and sensor quality.

Condition	PPO recurrent	greedy_unc	Difference (CI)
Budget 8 tiles	0.206	0.161	+0.044 (+0.036 to +0.053)
Budget 16 tiles (core)	0.268	0.225	+0.043 (+0.035 to +0.051)
Budget 32 tiles	0.312	0.289	+0.023 (+0.018 to +0.029)
Revisit 6 h / 24 h	0.262 / 0.263	0.222 / 0.219	-
Latency 1 pass	0.199	0.166	-
Sensor degraded / perfect	0.153 / 0.476	0.137 / 0.429	-

7.8 Real-event replays (Assam)

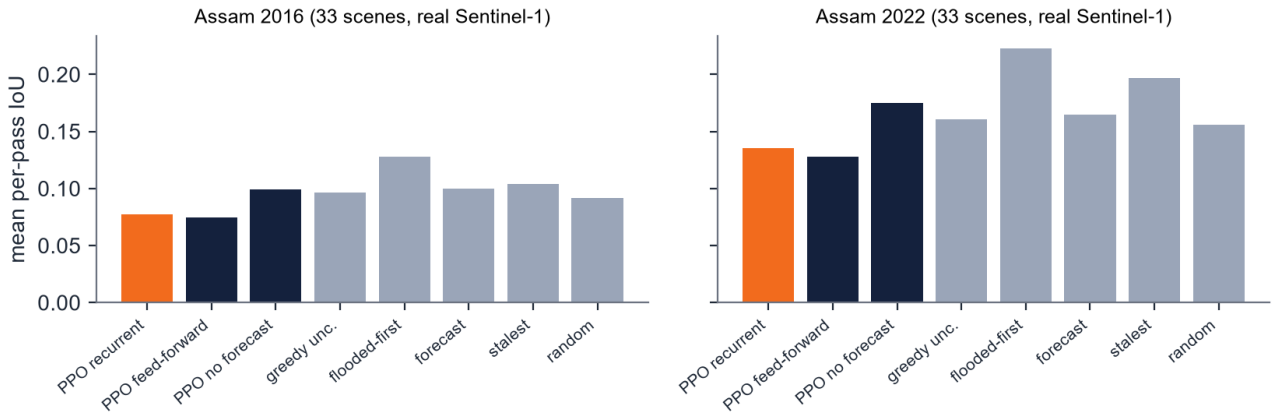


Figure 7.5 - Mean per-pass IoU on real Sentinel-1 replays (operational pseudo-forecast).

Event (33 scenes)	PPO recurrent	PPO no forecast	greedy_unc	flooded_first	stalest
Assam 2016 (4-8 passes)	0.077	0.099	0.096	0.128	0.104
Assam 2022 (16-25 passes)	0.135	0.175	0.160	0.223	0.197

The simulated advantage does not transfer to the replays. The reference masks themselves reach only IoU 0.337 against the hand labels on 2016-08-12 (the deployed U-Net 0.090 at 60 m), so reference noise is as large as the policy differences. The pseudo-forecast also hurts the forecast-aware agent: with forecasts marked missing, recurrent PPO improves to 0.106 (2016) and 0.165 (2022).

7.9 Flood outlook

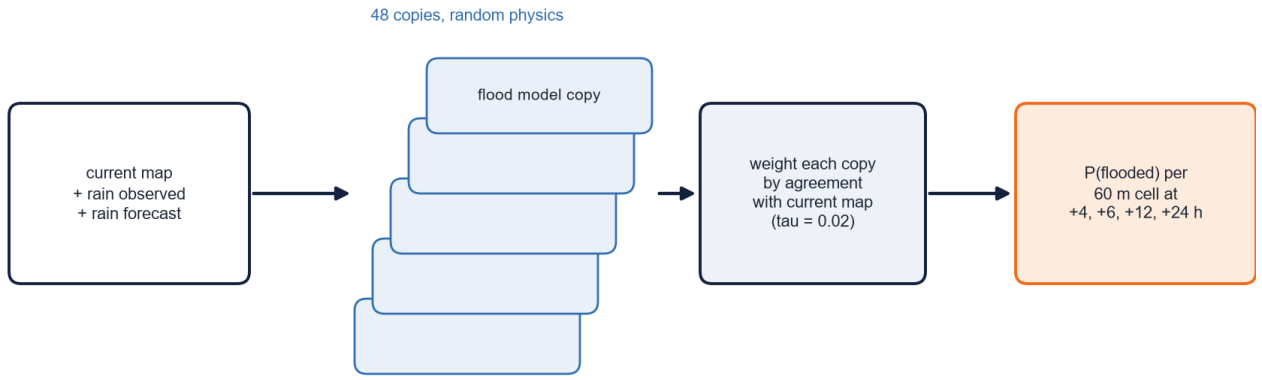


Figure 7.6 - Ensemble flood outlook with likelihood weighting.

Test	Outlook IoU	Persistence IoU	Evidence
Simulated test, +4 h	0.646	0.217	paired gain +0.43 (CI +0.39 to +0.47)
Simulated test, +6 h	0.655	0.215	+0.45 (+0.41 to +0.49)
Simulated test, +12 h	0.648	0.205	+0.44 (+0.40 to +0.47)
Simulated test, +24 h	0.675	0.194	+0.49 (+0.45 to +0.52)
Assam 2016, next real image	0.059	0.036	better in 63 % of 143 cases
Assam 2022, next real image	0.122	0.108	better in 59 % of 672 cases

Simulated scores are a best case because the truth comes from the same model; the unweighted ensemble is equally good there (+4 h IoU 0.671), so terrain carries most of the skill. On real data, the gain over persistence is small.

7.10 Computational cost

Measure	Recurrent	Feed-forward
Parameters	769,474	308,930
Latency per decision, 1 CPU thread (mean / p95)	5.9 / 9.1 ms	5.3 / 6.4 ms
16-tile plan	94 ms	84 ms
Training time per run	60-100 min	46-93 min
Peak memory (benchmark)	477 MB	-

7.11 Requirements

ID	Requirement	Achieved	Status
R1	Per-pass IoU reported	All policies, all passes	Met
R2	Final IoU ≥ best baseline; |area bias| ≤ 0.25	0.354 vs 0.303; bias +2.56 pooled, +0.12 India	Partly met
R3	Not below best baseline at 8/16/32	+0.044 / +0.043 / +0.023	Met
R4	Median passes to IoU 0.5 ≤ baseline	13 vs 13 (censored); 17.7 % vs 4.2 % reach it	Met (weak)
R5	0 infeasible; 100 % budget	0 in 6,080 episodes	Met
R6	Drop ≤ 10 % under each corruption	Missing rain + gauge: -11.0 %	Not met
R7	≥ best baseline on India; ≥ 2 of 3 events	+0.040; 3 of 3	Met
R8	< 50 ms, < 5 M params, < 2 h	5.9 ms, 0.77 M, 103 min	Met
R9	One-command regeneration; checksums	678 files, 0 mismatches	Met
R10	Non-reliance conditions	docs/operational_readiness.md	Met
R11	U-Net ≥ Otsu	0.652 vs 0.650	Met
R12	Reward validity ρ ≥ 0.7 (validation)	0.83 (test 0.57)	Met
R13	H1 with Holm p < 0.05 and CI excluding 0	5 of 7 comparisons	Partly met

8. Discussion

8.1 What the results show

Within the simulation, a learned tasking policy clearly outperforms every hand-built rule, including a greedy expected-error heuristic whose weights were tuned on validation scenes. The effect is large relative to its uncertainty (d_z = 1.11 against the strongest baseline), consistent across all training seeds, all three unseen events and all budgets, revisit intervals, latencies and sensor qualities tested. The learned policy also degrades gracefully under forecast corruption.

8.2 Why memory and forecasts did not help

The two components that name the method did not add measurable value. The following explanations are plausible but were not tested separately:

- The observation already contains the shared estimator's state, the observation age and the never-observed flag per tile. This summary may make the problem close to Markov for the agent, so an LSTM has little left to remember.
- The synthetic forecast is derived from the same simulated rain that also enters the observed-rain channel, the gauge and, indirectly, the current map. Its additional information may be small relative to these inputs.
- Recurrent PPO used hyperparameters tuned for the feed-forward agent; a recurrent-specific tuning was not done.
- The flood regime of the simulator may change too slowly within a 6-day episode for 24-72 h forecasts to change which tile is most valuable.

8.3 Simulation-to-reality gap

The result that matters most for practice, the real Assam replays, is negative. Contributing factors include (i) model-derived reference masks with IoU 0.337 against hand labels, (ii) a real flood regime with persistent monsoon flooding that the Gaussian-storm simulator represents only partly, (iii) much longer pass sequences in 2022 (up to 25) than the 12 seen in training, (iv) the RTC-to-GRD domain shift of the U-Net, and (v) the ERA5 pseudo-forecast. The observation that the forecast-free agent does better on real data than the forecast-aware one supports (v) as one factor.

8.4 Threats to validity

- Only three test events exist; event-level variance is poorly estimated, so generalisation claims stay qualitative.
- The sensor model is uniform over terrain and pessimistic on test events (simulated single-image IoU 0.57 versus about 0.70 for the real U-Net).
- The pooled area-bias criterion (R2) is unstable for small floods; the median bias is +0.019.
- Reward validity on the test set (ρ = 0.57) is lower than on validation (0.83).
- GPU training is not bit-reproducible, although evaluation of the saved checkpoints is.

9. Limitations and operational readiness

9.1 Limitations

Area	Limitation
Labels	No multi-date flood labels exist; flood dynamics are simulated; replay references are model-derived
Forecasts and gauges	No archived NWP; synthetic or ERA5 pseudo-forecasts; no real gauge series
Terrain	Surface DEM (vegetation and buildings raise HAND); 60 m resampling loses small channels and embankments
Flood model	Conceptual, not hydrodynamic: no momentum, backwater, levee breach, dam operation or pluvial ponding
Calibration	Matched to flooded-fraction distribution (KS 0.13), not to flood timing
Sensor	Parametric error model, uniform over land cover; U-Net only ties Otsu
Estimator	Keeps the latest observation; does not propagate dynamics into stale tiles
Tasking	Emulates agile commercial SAR; swaths are random bands, not orbits; tiles selected greedily within a pass
Licence	Sen1Floods11 licence is not stated in its repository; data are used for research and not redistributed

9.2 Operational readiness

The system is a research prototype. Before any operational trial with agencies such as ASDMA, NDMA or NRSC/ISRO, it would need real-time ingestion of SAR, forecasts and gauges; a user-facing uncertainty product with an explicit unknown class; tamper-evident audit logging; regional fine-tuning of the segmentation model on Indian labels; analyst approval of every tasking plan with a heuristic fallback; integration with a real operator's orbit and tasking API; and blind validation by an independent team.

CONDITIONS UNDER WHICH OUTPUTS MUST NOT BE RELIED ON

Outside lowland riverine floodplains (flash floods in steep terrain, coastal surge, urban pluvial flooding).

Where the segmentation model has not been validated for the region and sensor.

When forecasts or gauges are degraded, unless the corresponding robustness results have been checked.

For any life-safety decision without a human analyst reviewing the map and its uncertainty.

10. Conclusion and future work

10.1 Conclusion

This project built a complete, reproducible pipeline for learning flood-mapping satellite tasking: real SAR segmentation and terrain data, a calibrated conceptual flood simulator, a leakage-tested POMDP environment, a masked per-tile actor-critic trained with PPO, validation-tuned baselines, a frozen evaluation protocol and a real-event replay over Assam. In simulation, the learned policy improves the mean per-pass IoU of the flood map by 0.043 over the best heuristic (0.040 on Assam scenes), robustly across seeds, events and operating conditions, at a decision cost of about 6 ms. The forecast and memory components did not add value, and the advantage did not transfer to real Sentinel-1 replays. The evidence therefore supports the narrower claim that learned tasking beats hand-built tasking in simulation, and identifies the simulation-to-reality gap as the main open problem.

10.2 Future work

1. Train and validate on real multi-date flood observations (for example, dense Sentinel-1 series with expert labels for several Indian monsoon seasons).
2. Use archived operational forecasts (IMD or ECMWF) and real CWC gauge data where access can be arranged.
3. Train on variable and longer pass sequences that match real revisit patterns, and use real orbit propagation and agility limits.
4. Improve the estimator by propagating flood dynamics into stale tiles (for example with the outlook ensemble), and re-test the ranking of policies.
5. Tune the recurrent agent separately and test settings in which memory is necessary (for example, without per-tile age inputs).
6. Optimise the tile set of a pass jointly, and model downlink and processing latency explicitly.
7. Fine-tune the segmentation model on Indian flood labels and on RTC inputs.

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Appendix A. Configuration

configs/env.yaml (main entries)

```
passes: 12              pass_interval_h: 12      spinup_h: 48
budget_per_pass: 16     latency_passes: 0        eval_delay_h: 3
swath_tiles: [8, 16]    full_swath_prob: 0.25
acq_fail_prob: 0.05     nodata_tile_prob: 0.05
physics (after calibration): storm_total_mm [27.3, 216.0], runoff_coef [0.075, 0.175],
         ext_scale [0.125, 2.0], residence_h [12, 72], ext_lag_h [12, 60], loss_mm [2.5, 7.5]
forecast: mode operational, gauge/rain-obs missing 0.1, forecast missing 0.05
estimator: hand_prior     reward: err, fn_weight 3.0, unc_coef 0.0
```

configs/ppo.yaml

```
num_envs: 16   total_steps: 1500000   lr: 0.001   gamma: 0.9   gae_lambda: 0.95
clip: 0.2      ent_coef: 0.003        vf_coef: 0.5   epochs: 8    minibatches: 4
max_grad_norm: 0.5   eval_every: 25   n_val_episodes: 32
```

Appendix B. Evidence files

Result	File
Core and India tables	results/analysis/core_table_all_test_events.csv, core_table_India.csv
Statistics	results/analysis/stats_core_all_test_events.json, stats_core_India.json, stats_conditions.json
Robustness	results/analysis/conditions_mean_iou.csv
Episodes and audit trail	results/experiments/<suite>/episodes.jsonl
Training logs and checkpoints	results/rl/<variant>/seed<n>/train_log.json, best.pt
Segmentation	results/segmentation/eval_test.json
Sensor model, calibration, prior	results/sensor_model.json, sim_calibration.json, hand_prior.json
Reward validity, PPO sweep	results/reward_validity_val.json, ppo_sweep.json
Latency	results/latency_bench.json
Outlook	results/outlook_val.json, outlook_test.json, outlook_real.json
Protocol freeze, manifest	results/protocol_freeze.json, results/data_manifest.json

Appendix C. Example episode

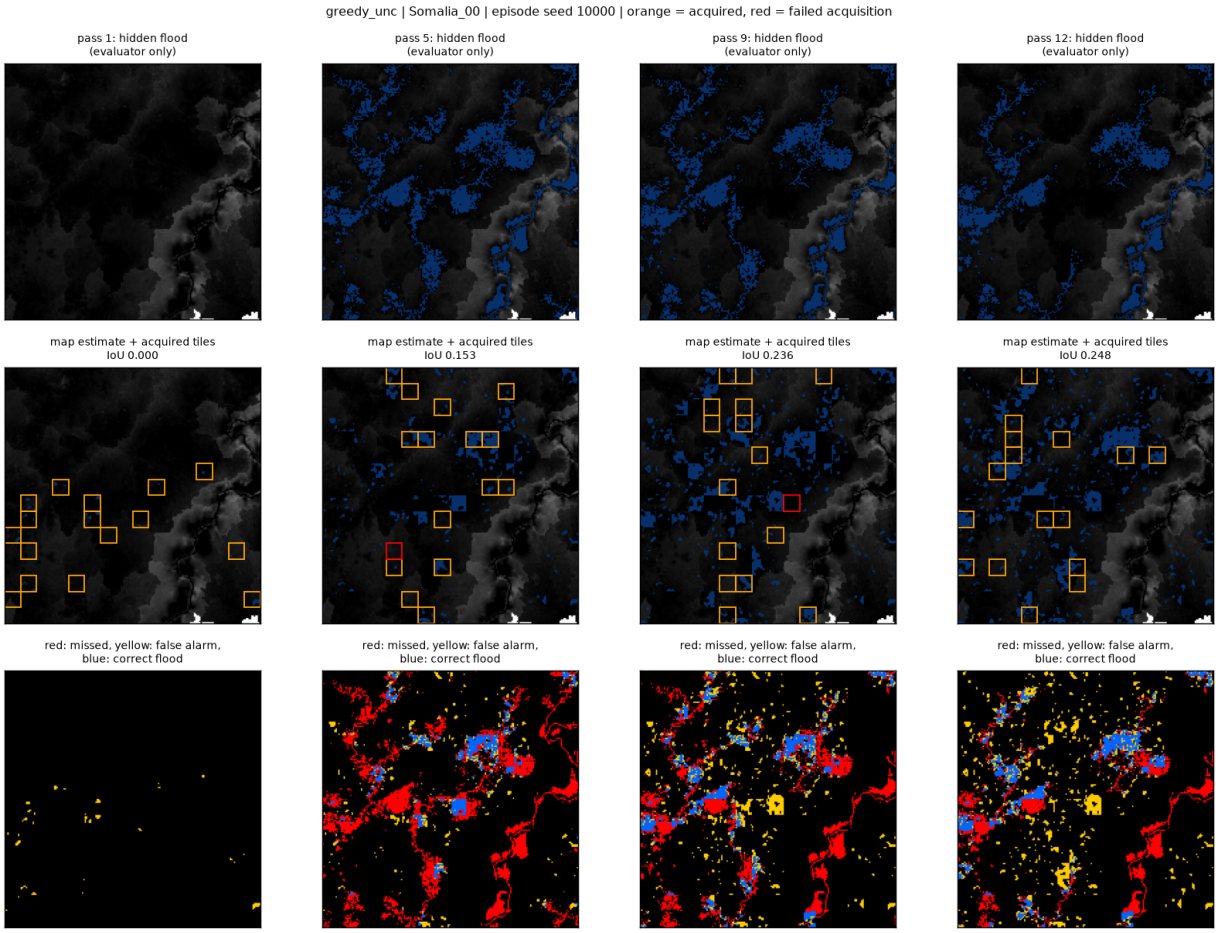


Figure C.1 - Validation episode of greedy_unc on Somalia_00: hidden flood, map with tasked tiles, and error map at passes 1, 5, 9 and 12.